

Hallucination Failure Types

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of generation failures and verification.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Multimodal Evidence. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Structured Output and Schemas. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 14: Hallucination Failure Types

Fluent output may contain unsupported claims, fabricated detail, or faulty inference.



CREATOR MISSION

Tag claims in an answer and classify each failure type.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Controlled prompt experiments. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week's end goal

By the end, I can distinguish supported, contradicted and unsupported generated claims.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use hallucination failure types to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Generated answers may invent facts, misread quantities or cite irrelevant passages. Verification checks each claim against evidence rather than accepting fluent wording.

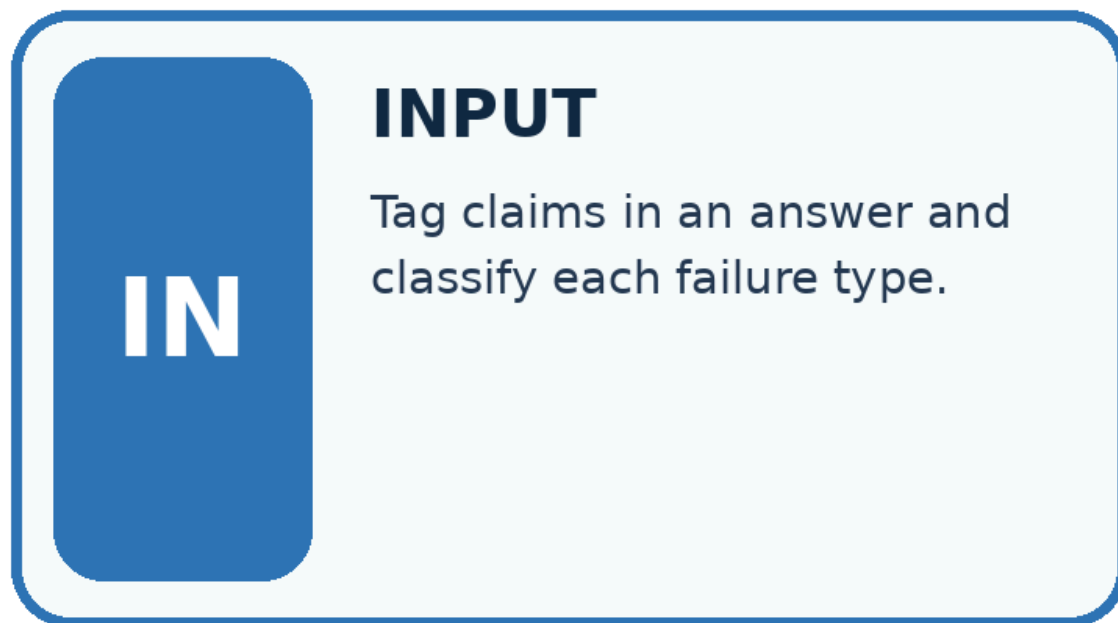
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Fluent output may contain unsupported claims, fabricated detail, or faulty inference.



OUT

EVIDENCE

Tag claims in an answer and classify each failure type.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Split an answer into atomic claims. For each, record the exact source passage, support status and any needed calculation. Mark unsupported and contradicted claims separately; do not repair them by inventing sources.

Generation failures and verification — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Source R says room B seats 12 and the meeting starts at 15:00. Generated answer says B seats 20, starts at 15:00 and includes free lunch.

Check or calculation: Split an answer into atomic claims. For each, record the exact source passage, support status and any needed calculation. Mark unsupported and contradicted claims separately; do not repair them by inventing sources.

Conclusion and limit: Capacity 20 is contradicted; 15:00 is supported; free lunch is unsupported. One of three claims has support. The missing lunch fact is not evidence of no lunch.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Add a planned 30-minute stop while preserving speed and distance.

Prediction: Planned total elapsed time becomes 3.5 hours.

Reason: Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

FABRICATION

invented detail or source

3

FAULTY INFERENCE

conclusion not justified by
evidence

4

CITATION ERROR

source does not support the claim

Word · My own example · Not this / boundary

Word: unsupported claim

My own example: A statement for which the supplied evidence does not establish the asserted conclusion.

Not this / boundary: _____

Word: fabrication

My own example: Invented information presented as if it were established fact.

Not this / boundary: _____

Word: faulty inference

My own example: A conclusion that does not follow from the available premises or evidence.

Not this / boundary: _____

Word: citation error

My own example: A reference that is wrong, missing or does not support the claim attached to it.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Atomic claim

Meaning: A single factual statement checked separately

Word: Contradicted

Meaning: Inconsistent with the supplied evidence

Word: Unsupported

Meaning: Not established by the available evidence

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 14 • WORKED EVIDENCE

- 1 Claim 1 is directly supported by the passage.
- 2 Claim 2 invents a scientist's quotation: fabrication.
- 3 Claim 3 draws a cause from correlation: faulty inference.
- 4 Tag each claim, repair only with evidence, or mark unknown.

1. Claim 1 is directly supported by the passage.
2. Claim 2 invents a scientist's quotation: fabrication.
3. Claim 3 draws a cause from correlation: faulty inference.

4. Tag each claim, repair only with evidence, or mark unknown.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Source R says room B seats 12 and the meeting starts at 15:00. Generated answer says B seats 20, starts at 15:00 and includes free lunch.

Capacity 20 is contradicted; 15:00 is supported; free lunch is unsupported. One of three claims has support. The missing lunch fact is not evidence of no lunch.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Hallucination Failure Types

A fictional Study Buddy drafts a field-trip summary from one approved record.

The facts we will use

Source S: trail length 12 km; planned constant speed 4 km/h, excluding stops; departure 09:00.

Draft: “The walk takes two hours, starts at 09:00 and includes free lunch.” No lunch record is supplied.

Hallucination Failure Types

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute duration

$12 / 4 = 3$ hours, excluding stops.



2. Contradicted claim

Two-hour duration disagrees with the calculation from S.



3. Supported claim

Departure 09:00 is stated in S.



4. Unsupported claim

Free lunch has no source; missing evidence is not a confirmed no-lunch fact.

CHANGE ONE THING

Add a planned 30-minute stop while preserving speed and distance.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Compute duration

What happens and why: $12 \div 4 = 3$ hours, excluding stops.

Step: 2. Contradicted claim

What happens and why: Two-hour duration disagrees with the calculation from S.

Step: 3. Supported claim

What happens and why: Departure 09:00 is stated in S.

Step: 4. Unsupported claim

What happens and why: Free lunch has no source; missing evidence is not a confirmed no-lunch fact.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Different errors need different repairs. Recalculate the duration; keep the supported departure; remove or seek evidence for lunch. These categories are more informative than calling the whole answer wrong. Arithmetic inherits assumptions such as constant speed and no stops, so avoid turning the three-hour walking calculation into a guaranteed arrival time.

What this example does not establish

The fictional plan does not establish actual speed, delays or what eventually occurred on a real trip.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A confident, detailed answer is less likely to hallucinate. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Source agreement and arithmetic validity are separate checks; missing evidence is not proof of the opposite claim.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Generation failures and verification”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Generation failures and verification. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Can we conclude there will definitely be no lunch from S alone?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Color-code every claim in a generated paragraph, classify failures, attach supporting passage lines, and rewrite with honest uncertainty.

Color-code every claim in a generated paragraph, classify failures, attach supporting passage lines, and rewrite with honest uncertainty.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY
1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Study Buddy must distinguish source-backed claims, calculations, and unverified generation.

THIS WEEK’S CHECKPOINT Study Buddy must distinguish source-backed claims, calculations, and unverified generation.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

Audit Study Buddy outputs claim by claim, recording contradiction, missing support and correct calculations separately.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Tag claims in an answer and classify each failure type.

CHALLENGE Tag claims in an answer and classify each failure type.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 14 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

An answer invents a quotation, cites a source lacking it and supplies an impossible date.

The question

Separate the failures and checks.

Worked explanation

Check exact quote text, claim-to-source support and calendar consistency independently. Each failure needs its own correction.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Different failures require different checks

Evidence or stage · Value or result

Evidence or stage: Quotation

Value or result: Locate exact source text

Evidence or stage: Citation

Value or result: Check support for the claim

Evidence or stage: Date

Value or result: Check source and calendar consistency

Evidence or stage: Repair

Value or result: Correct or remove unsupported statements

An answer invents a quotation, cites a source lacking it and supplies an impossible date.

Separate the failures and checks.

Check exact quote text, claim-to-source support and calendar consistency independently. Each failure needs its own correction.

14. Try a new case independently

Independent case: Apply a new case independently

Source S: trip is 18 km; vehicle speed is 6 km/h, constant, excluding stops. Answer says 2 hours, 18 km and includes lunch. Audit and recalculate.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 14 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An answer gives a book title that does not exist and quotes words absent from the cited passage. Identify two failures and a repair.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Study Buddy must distinguish source-backed claims, calculations, and unverified generation.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A generated answer contains a nonexistent quote, a real source unrelated to the claim and a correct sum using the wrong input. Classify the three failures and their repairs.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.